GCSE · OCR · Computer Science

2 hours 29 questions

Computational Thinking, Searching & Sorting Algorithms

Computational Thinking, Searching & Sorting Algorithms

Principles of Computational Thinking / Standard Searching Algorithms / Standard Sorting Algorithms

Easy (5 questions)	/12
Medium (15 questions)	/73
Hard (9 questions)	/33
Total Marks	**/118**

Scan here to return to the course
or visit savemyexams.com

© 2026 Save My Exams, Ltd.

Easy Questions

1 **State** the **name** of each of the following **computational thinking techniques**.

Breaking a complex problem down into smaller problems.

Hiding or removing irrelevant details from a problem to reduce the complexity.

.....

..... **(2 marks)**

2 The following table contains several definitions of terms that are used in Computer Science.

Letter	Definition
A	Cleaning up data entered by removing non-standard characters
B	Hiding or removing irrelevant details from a problem to reduce complexity
C	Checking that the user is allowed to access the program
D	Breaking a complex problem down into smaller problems
E	Repeating elements of a program
F	Converting one data type to another, for example converting an integer to a real number

Write the letter of the **definition** that **matches each keyword** in each space.

© 2026 Save My Exams, Ltd.

Keyword	Letter
Decomposition	
Abstraction	
Input sanitisation	
Casting	

.....

.....

.....

..... **(4 marks)**

3 (a) A list of valid **discount codes** is shown below.

[NIC12B, LOR11S, STU12M, VIC08E, KEI99M, WES56O, DAN34S]

State one reason why a **binary search** would not be able to be used with this data.

..... **(1 mark)**

(b) **Give the name** of **one searching algorithm** that would be able to be used with this data.

..... **(1 mark)**

4 (a) State **one** pre-requisite for a binary search algorithm.

.....**(1 mark)**

(b) **Tick (✓)** one box to identify the name of the sorting algorithm that splits data into individual items before recombining in order.

A. Bubble sort

B. Insertion sort

C. Merge sort

(1 mark)

© 2026 Save My Exams, Ltd.

5 (a) Students take part in a sports day. The students are put into teams.

Students gain points depending on their result and the year group they are in. The points are added to the team score.

The team with the most points at the end of the sports day wins.

Abstraction and decomposition have been used in the design of the sports day program.

Identify one way that abstraction has been used in the design of this program.

..... **(1 mark)**

(b) Identify **one** way that decomposition has been used in the design of this program.

..... **(1 mark)**

© 2026 Save My Exams, Ltd.

Medium Questions

1 (a) A program stores the following list of positive and negative numbers. The numbers **need sorting** into **ascending orde**r using a **merge sort**.

45	**12**	**-99**	**100**	**-13**	**0**	**17**	**-27**

The first step is to divide the list into individual lists of one number each. **This has been done for you**.

Complete the merge sort of the data by **showing each step** of the process.

45	12	–99	100	–13	0	17	–27

(3 marks)

(b) Once the numbers are in order, a binary search can be run on the data.

Describe the steps a binary search will follow to look for a number in a sorted list.

.....

.....

.....

..... **(4 marks)**

(c) A linear search could be used instead of a binary search.

Describe the steps a linear search would follow when searching for a number **that is not** in the given list.

.....

..... **(2 marks)**

2 A sample of data is shown in **Fig. 4.**

amber	**house**	**kick**	**moose**	**orange**	**range**	**wind**	**zebra**

Fig. 4

Show the **stages of a binary search** to find the word **zebra** using the data shown in **Fig. 4**.

.....

.....

.....

..... **(4 marks)**

3 (a) **Tick (✓) one box in each row** to identify whether each statement about the **insertion sort** is true or false.

Statement	True (✓)	False (✓)
The list of words is initially split into a sorted set and an unsorted set.		
The insertion sort uses a divide stage and then a conquer stage.		
The list of words must be in order before the insertion sort can start.		
Each word is inserted into the correct place in the array, one by one.		
The insertion sort will not work because the word "wall" appears twice.		

(5 marks)

(b) The **sorted list** of words is shown below.

flour	**house**	**pumpkin**	**wall**	**wall**

Explain how a binary search would be used to try to find whether the word "**house**" appears in this list.

.....

.....

.....

..... **(4 marks)**

4 Taylor has used **computational thinking techniques** to develop an algorithm.

Give two computational thinking techniques that Taylor could have used, and **describe how**.

.....

.....

.....

..... **(4 marks)**

5 The following names of students are stored in an **array** with the **identifier** studentnames.

studentnames = ["Rob", "Anna", "Huw", "Emma", "Patrice", "Iqbal"]

Describe the steps that a **linear search** would take to find Anna in studentnames

.....

.....

.....

..... **(4 marks)**

6 The names of students are **sorted** into **ascending alphabetical order** using an **insertion sort**.

Complete the following diagram to **show the stages an insertion sort** would take to complete this task.

Each **row represents one pass** of the insertion sort algorithm.

You **may not need** to use all empty rows.

Rob	Anna	Huw	Emma	Patrice	Iqbal

(5 marks)

7 (a) Elliott plays football for OCR FC.

He wants to create a program to **store the results of each football match** they play and **the names of the goal scorers**.

Elliott wants individual players from the team to be able to submit this information.

Define what is meant by **abstraction**.

.....

..... **(2 marks)**

(b) **Give one example** of how abstraction could be used when developing this program in **part (a)**.

..... **(1 mark)**

8 (a) A library sorts their books based on a book code.

Show the steps that a **merge sort** would take to put the following list of book codes into **ascending alphabetical order** (from A to Z).

POE12 , BAC97 , FLY77 , JAV16 , TAL86 , AND18 , ZAR09 , HOP86

.....

.....

.....

..... **(4 marks)**

(b) **Explain one advantage** of a **merge sort** compared to a **bubble sort**.

.....

..... **(2 marks)**

9 **Show** how a binary search will be used to find the number 10 in the following data set:

1 2 5 6 7 10 20

.....

.....

..... **(3 marks)**

10 The student names for a team are stored in an array with the identifier theTeam

An example of the data in this array is shown:

Index	0	1	2	3	4	5
Data	Ali	Eve	Ling	Nina	Sarah	Tom

theTeam

A **linear search** function is used to find whether a student is in the team. The function:

- takes a **student name** as a parameter
- returns **True** if the student name is in the array
- returns **False** if the student name is not in the array

Complete the design of an algorithm for the linear search function.

```
function linearSearch(studentName)

   for count = 0 to ......................................

      if theTeam[.....................................] == ..................................... then

         return ......................................

      endif

   next count

   return False

endfunction
```

.....

.....

.....

..... **(4 marks)**

11 A list of numbers is stored in the following order:

42, 17, 8, 31, 25, 14

Describe how the list would be sorted into **ascending order** using an **insertion sort**

You must show the list **after passes 1, 2, 3 and 4**, and then give the **final sorted list**

(5 marks)

12 A list of words is stored in the following order:

grape, apple, cherry, banana, fig, date

Show how the list would be sorted into **alphabetical order** using an **insertion sort**

You must show the list **after passes 1, 2, 3 and 4**, and then give the **final sorted list**

(5 marks)

13 A warehouse organises its inventory using product codes.

Show the steps that a **merge sort** would take to put the following list of product codes into **ascending numerical order** (from smallest to largest).

847 , 215 , 632 , 419 , 756 , 128 , 903 , 541

.....

.....

.....

..... **(4 marks)**

14 A library organises its book collection using reference numbers.

Show the steps that a **bubble sort** would take to put the following list of reference numbers into **ascending numerical order** (from smallest to largest).

523 , 187 , 412 , 695 , 234 , 876 , 341 , 158

.....

.....

.....

..... **(4 marks)**

15 A sports club records the scores of its members using player IDs.

Describe the steps that a **bubble sort** would take to put a list of player IDs into **ascending numerical order** (from smallest to largest).

742 , 315 , 628 , 891 , 456 , 203 , 567 , 134

.....

.....

.....

..... **(4 marks)**

© 2026 Save My Exams, Ltd.

Hard Questions

1 A program uses a file to store a list of words that can be used in a game.

A sample of this data is shown in **Fig. 3**.

crime	bait	fright	victory	nibble	loose

Fig. 3

Show the stages of a bubble sort when applied to data shown in **Fig. 3**.

.....

.....

.....

..... **(4 marks)**

2 An **insertion sor**t algorithm is used to sort a dataset of game scores from smallest to largest

game_scores = [101, 110, 130, 50, 66]

To perform an insertion sort, a **count-controlled loop** must be used.

Describe the purpose of the count-controlled loop and give an **example** of what the loop would look like.

Purpose	
Example	

.....

.....

..... **(3 marks)**

3 **Describe** the steps involved in performing a **bubble sort** on the list in Fig. 1. **Show the state** of the list after each complete pass through the list.

12	34	45	23	8	39

Fig. 1

.....

.....

.....

.....

..... **(5 marks)**

4 **Explain** why the removal of geographical accuracy in a city bus map is a necessary form of abstraction for a computer

.....

.....

..... **(3 marks)**

5 **Justify** the use of decomposition in a large-scale banking app, focusing on independent testing

.....

.....

..... **(3 marks)**

6 **Analyse** how abstraction and decomposition help when designing an algorithm to solve a problem

.....

.....

.....

..... **(4 marks)**

7 A dataset of 1,000,000 items is **unsorted**.

Evaluate the effectiveness of using a binary search in this scenario

.....

.....

.....

..... **(4 marks)**

8 **Explain** a situation where a linear search would be more efficient than a binary search, even when the list is sorted

.....

.....

..... **(3 marks)**

9 **Explain** why a merge sort is often preferred over a bubble sort when sorting a very large list, such as 10 million website users

.....

.....

.....

..... **(4 marks)**

© 2026 Save My Exams, Ltd.